

Notes: Kehrig and Vincent (RES, 2026)

Good Dispersion, Bad Dispersion

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1 Big picture

- **Question.** Does dispersion in marginal revenue products always indicate misallocation, or can some observed dispersion reflect *better* resource allocation inside firms?
- **Main empirical fact.** In U.S. manufacturing, most dispersion in marginal revenue products of capital and most dispersion in investment rates occur *within* firms rather than *between* firms. The paper estimates that about 60% of *mrpk* dispersion and 68% of investment-rate dispersion come from differences across plants owned by the same firm.
- **Central challenge to the standard view.** In the Hsieh–Klenow tradition, more dispersion usually means bigger wedges and lower output. Kehrig and Vincent show that this monotonic interpretation can fail once firms use internal capital markets to reallocate funds across plants.
- **Main mechanism.** A multi-plant firm faces plant-level fixed and convex investment adjustment costs, plus firm-level costly external finance. If internal capital markets work well, the firm economizes on costly external finance by *staggering* investment across plants over time. That staggering raises within-firm dispersion in investment rates and expected *mrpk*, but it also raises aggregate output.
- **Headline quantitative result.** Moving from frictionless internal capital markets to no internal capital markets lowers aggregate capital by about 12% and output by about 8%, but it also lowers within-firm expected *mrpk* dispersion by more than 40%.
- **Why “good dispersion” matters.** In this environment, a reduction in frictions can increase both output and dispersion. The observed dispersion is then “good” because it is generated by a more efficient second-best allocation rather than by larger distortions.
- **Empirical support for the mechanism.** Firms whose plants are geographically closer together — so headquarters can plausibly gather information and move funds more easily — exhibit higher within-firm dispersion, lower average *mrpk*, less external finance, and investment dynamics more consistent with staggered plant-level spikes.
- **Broader implication.** Standard misallocation comparisons between the U.S. and poorer economies may understate true gaps because developed economies contain more multi-plant firms and therefore more scope for good dispersion. In the paper’s illustrative exercises, the implied output gain for India rises from 47% to 54.5%, and for China from 39% to 57.4%.
- **Bottom line.** Dispersion is not a sufficient statistic for misallocation. One must separate *bad dispersion* coming from distortions from *good dispersion* generated by optimal within-firm reallocation under constraints.

2 Data and measurement (institutional and quantitative objects)

2.1 U.S. manufacturing data

- **Main source.** The paper uses plant-level U.S. Census manufacturing data, primarily the Annual Survey of Manufactures (ASM), supplemented by the Census of Manufactures and auxiliary Census datasets.

- **Unit of observation.** The Census establishment is a physical production unit at a single location. The paper refers to establishments as *plants*. A *firm* is the collection of plants under common ownership or control.
- **Why plants versus firms matters.** Multi-plant firms operate only about 28% of plants, but they account for roughly 80% of aggregate manufacturing activity. So understanding within-firm reallocation is quantitatively essential.

2.2 Measurement of marginal revenue products and investment

- **Production-side approximation.** Assuming a common Cobb–Douglas technology within 4-digit NAICS industries, the paper approximates the marginal revenue product of capital by real value added per unit of capital:

$$\text{mrpk}_{nt} \approx \frac{y_{nt}}{k_{nt}}, \quad \log \text{mrpk}_{nt} = \log y_{nt} - \log k_{nt}.$$

- **Dispersion object.** The key empirical statistic is the cross-sectional variance of $\log \text{mrpk}_{nt}$ within 4-digit industries.
- **Investment object.** Plant investment intensity is measured by the investment rate i_{nt}/k_{nt} .

2.3 Variance decomposition: within-firm versus between-firm dispersion

Within a 4-digit NAICS industry, total *mrpk* dispersion is decomposed as

$$\begin{aligned} V_t(\text{mrpk}_{nt}) &= \sum_j \omega_{jt} (\text{mrpk}_{jt} - \text{mrpk}_t)^2 + \sum_j \omega_{jt} \sum_{n \in j} \omega_{nt}^j (\text{mrpk}_{njt} - \text{mrpk}_{jt})^2 \\ &= V_t^B + V_t^W. \end{aligned} \tag{1}$$

- **Interpretation.** V_t^B is between-firm dispersion. V_t^W is the average within-firm dispersion across plants.
- **Main finding.** In the full sample, about 60.1% of *mrpk* dispersion and 67.9% of investment-rate dispersion are within-firm. Even in narrow, homogeneous-product industries, the within-firm shares remain large at about 55.9% and 68.9%.
- **Economic meaning.** The allocative problem is not mainly across independent firms; it is largely inside multi-plant firms.

2.4 Descriptive cross-sectional moments

- **Dispersion of *mrpk*.** The standard deviation of $\log \text{mrpk}$ is 0.905. A one-standard-deviation-higher plant therefore produces about $e^{0.905} \approx 2.5$ times the value added of the average plant, holding capital fixed.
- **Tail spread.** The interdecile range of $\log \text{mrpk}$ is 2.120, implying roughly an $e^{2.12} \approx 8.3$ -fold difference in value added per unit of capital between the 90th and 10th percentiles.
- **Skewness.** The distribution is positively skewed, with conventional skewness 0.634 and Kelley skewness 0.128.

- **Investment lumpiness.** The average investment rate is 0.112, while the cross-sectional standard deviation is 0.362 and excess kurtosis is 57.2. This is the data-side motivation for fixed adjustment costs.

2.5 Geographic distance as a proxy for internal-capital-market frictions

- **Proxy.** The paper uses the log of one plus the average distance of plants from the firm’s geographic center point as a proxy for frictions to internal capital markets.
- **Logic.** More distance means headquarters must spend more time and resources gathering information, evaluating projects, and redirecting funds.
- **Construction.** Using latitude and longitude, the authors compute a value-added-weighted center for each firm and then compute the value-added-weighted average distance of plants from that center.
- **Summary statistics.** The mean distance measure is 4.70, the standard deviation is 2.34, and skewness is -0.96 . The mean corresponds to plants being about 55 km from the firm center; one standard deviation above the mean corresponds to about 571 km.

2.6 Validation and cross-country ingredients

- **Non-targeted validation moment.** After removing plant fixed effects, the empirical within-firm variance of *mrpk* for two-plant firms is 0.214, against 0.492 in raw data. The model-implied range, 0.149 to 0.255 as γ varies, lines up reasonably well.
- **Cross-country extension.** For India and China, the paper uses Hsieh–Klenow-style dispersion accounting in revenue productivity. The point is not to build a new plant dataset for those countries, but to show how ignoring within-firm good dispersion can understate misallocation gaps.

3 Models and empirical specifications (core equations + interpretation)

3.1 Conceptual framework: when dispersion can be good

The paper begins with a stripped-down example. A firm allocates funds across N plants. Each plant has capital k , each can receive investment i , and there is a fixed investment adjustment cost τ :

$$y = \begin{cases} k^\alpha, & \text{if no investment,} \\ (k + i - \tau)^\alpha, & \text{if investment,} \end{cases} \quad 0 < \alpha < 1. \quad (2)$$

If the firm invests in a fraction n^* of plants and splits funds equally across them, dispersion in log *mrpk* becomes

$$n^*(1 - n^*)[\alpha \log(1 + i^*/k)]^2 > 0. \quad (3)$$

- **Bad dispersion.** Starting from no adjustment cost, raising τ pushes the firm away from the frictionless optimum, raises dispersion, and lowers output.

- **Good dispersion.** Starting from such high frictions that the firm does not invest at all, lowering τ induces investment in a subset of plants, raises output, and also raises dispersion.
- **Internal-capital-market version.** If plant-level funds cannot be pooled, no plant may invest and dispersion is zero. Restoring the internal capital market lets the firm pool funds, invest in selected plants, and raise both output and dispersion.
- **Economic lesson.** The mapping from frictions to dispersion is non-monotone. Dispersion alone is not a sufficient statistic for misallocation.

3.2 Plant technology and shock processes in the quantitative model

Each plant $n \in \{A, B\}$ in firm j produces

$$y_{nt} = e^{z_{nt}} k_{nt}^\alpha. \quad (4)$$

Productivity has a plant-specific and a firm-wide component:

$$z_{njt} = \rho_p z_{njt-1} + \eta_{njt}, \quad (5)$$

$$z_{jt} = \rho_f z_{jt-1} + \eta_{jt}. \quad (6)$$

Capital evolves as usual:

$$k_{nt+1} = (1 - \delta)k_{nt} + i_{nt}. \quad (7)$$

- **Interpretation.** The firm-wide TFP component makes investment incentives correlated across plants, which is exactly why synchronized spikes are likely in the absence of a good internal capital market.
- **Why only two plants?** The authors restrict the model to two plants to keep the non-convex dynamic programming problem numerically tractable while preserving the core economics of internal reallocation.

3.3 Plant-level adjustment costs and plant cash flow

Investment at the plant level faces both non-convex and convex adjustment costs:

$$\theta(i_{nt}, k_{nt}) = \left[\psi_1 I \left\{ \frac{i_{nt}}{k_{nt}} > \vartheta \right\} + \frac{\psi_2}{2} \left(\frac{i_{nt}}{k_{nt}} \right)^2 \right] k_{nt}. \quad (8)$$

Plant cash flow is therefore

$$\pi_{nt} = e^{z_{nt}} k_{nt}^\alpha - \theta(i_{nt}, k_{nt}). \quad (9)$$

- ψ_1 . Fixed cost of investing. This generates lumpy plant-level investment and makes investment spikes attractive relative to smooth adjustment.
- ψ_2 . Quadratic cost. This makes very large spikes more expensive at the intensive margin.
- **Why these costs matter.** Without fixed costs, the firm would tend to equalize marginal products more tightly. The good-dispersion mechanism relies on the interaction of lumpy investment with costly finance.

3.4 Financing under full internal capital markets

If the internal capital market is frictionless, the firm pools plant cash flows:

$$\Pi_t = \pi_{At} + \pi_{Bt}. \quad (10)$$

Firm investment must satisfy

$$i_{At} + i_{Bt} = I_t \leq \Pi_t + \phi_t, \quad (11)$$

where ϕ_t is external finance. Issuing external finance is costly:

$$\Phi_t^{\text{FullICM}} = [\xi_1 I\{\phi_t > 0\} + \xi_2 \max\{0, \phi_t\}] K_t. \quad (12)$$

- **Interpretation.** The firm can use cash generated at one plant to finance investment at the other. This is the benchmark in which internal capital markets operate perfectly.
- **Role of external-finance costs.** Because issuing equity is costly, the firm wants to smooth firm-wide financing needs even if it does *not* want to smooth plant-level investment.

3.5 Financing with no internal capital markets

At the other extreme, a plant can only finance its own investment:

$$i_{nt} \leq \pi_{nt} + \phi_{nt}, \quad n \in \{A, B\}. \quad (13)$$

The cost of external finance becomes

$$\Phi_t^{\text{NoICM}} = [\xi_1 I\{\phi_{At} \text{ or } \phi_{Bt} > 0\} + \xi_2 \max\{0, \phi_{At}\} + \xi_2 \max\{0, \phi_{Bt}\}] K_t. \quad (14)$$

The paper then nests the two polar cases with an internal-friction parameter γ :

$$\Phi_t(\phi_{At}, \phi_{Bt}, \gamma) = (1 - \gamma)\Phi_t^{\text{FullICM}} + \gamma\Phi_t^{\text{NoICM}}, \quad \gamma \in [0, 1]. \quad (15)$$

- **Interpretation of γ .** It is the probability-like weight on the high-friction, no-pooling regime. Higher γ means internal capital markets are less functional.
- **What changes when γ falls.** The firm can economize on external finance by rotating investment spikes across plants over time.

3.6 Firm problem and the core trade-off

Let $Z_t = \{z_{At}, z_{Bt}\}$ and $K_t = \{k_{At}, k_{Bt}\}$. The firm's value function is

$$V(Z_t, K_t) = \max_{i_{At}, i_{Bt}} \left\{ \Pi_t - I_t - \Phi_t + \beta \mathbb{E}[V(Z_{t+1}, K_{t+1})] \right\}.$$

- **Core tension.** Fixed plant-level adjustment costs make investment lumpy, but costly external finance makes synchronized spikes expensive.
- **Second-best response.** A firm with a functional internal capital market shifts funds across plants and *stages* investment projects through time. That policy raises dispersion of investment and expected *mrpk* while lowering financing costs.
- **This is the source of good dispersion.** More dispersion appears precisely because the firm uses internal reallocation to relax a binding financing friction.

3.7 Quantitative predictions from the model

- **Aggregate effect of shutting down internal capital markets.** Going from $\gamma = 0$ to $\gamma = 1$ lowers capital by about 12% and output by about 8%.
- **Dispersion effect.** The same move lowers within-firm expected *mrpk* dispersion by more than 40%.
- **Mechanism details.** With no internal capital markets, synchronized investment spikes occur about 9% of the time. With frictionless internal capital markets, synchronized spikes disappear and rotating spikes dominate.
- **Finance details.** The expected need for external funds is near zero in the FullICM economy but about 6% of capital in the NoICM economy. The average financing cost rises from less than 0.5% of capital to about 3.5%.

3.8 Development-accounting extension

To revisit Hsieh–Klenow, the paper starts from the conventional gain from reducing misallocation in an emerging economy (EM) to the U.S. benchmark:

$$\log(1 + \text{conv. output gain}) = \frac{1}{\sigma - 1} \log\left(\frac{\sum_i A_{n,US}^{\sigma-1}}{\sum_n A_{n,EM}^{\sigma-1}}\right) - \frac{\sigma}{2} [V(tfpr_{n,US}) - V(tfpr_{n,EM})]. \quad (16)$$

To separate multi-plant from single-plant contributions, they decompose total variance as

$$V(tfpr_n) = \omega^M V(tfpr_n^M) + (1 - \omega^M) V(tfpr_n^S) + \omega^M (1 - \omega^M) (\overline{tfpr}^M - \overline{tfpr}^S)^2. \quad (17)$$

- **Interpretation.** If the U.S. has more multi-plant firms, part of its measured dispersion is good dispersion coming from within-firm reallocation. Then matching U.S. dispersion is an overly weak benchmark for India or China.
- **Illustrative conclusion.** Once internal-capital-market-induced good dispersion is accounted for, the output gains from eliminating bad dispersion in emerging economies are larger than in the standard benchmark.

4 Identification, calibration, and credibility

4.1 How the quantitative model is disciplined

- **Not a causal-IV design.** The main model is identified by calibration and by matching plant- and firm-level moments rather than by quasi-experimental variation.
- **Production side.** $\alpha = 0.627$ is estimated by extending the structural framework of Cooper and Haltiwanger to multi-plant firms.
- **Capital law of motion.** $\delta = 0.067$ is chosen to match the average investment rate.
- **Shock processes.** $(\rho_p, \rho_f, \sigma_p, \sigma_f) = (0.60, 0.85, 0.25, 0.24)$ are chosen to match serial correlation and volatility of *mrpk* at plant and firm levels.

- **Adjustment costs.** $(\psi_1, \psi_2) = (0.039, 0.049)$ are taken from the plant-level investment literature.
- **External-finance costs.** $(\xi_1, \xi_2) = (0.00154, 0.11)$ are disciplined using single-unit firms so that the calibration of external finance is not contaminated by internal-capital-market frictions.

4.2 What gives the empirical section credibility

- **Clear empirical target.** The model predicts several margins at once: within-firm dispersion, the level of firm *mrpk*, external-finance use, synchronized spikes, and within-firm co-movement of investment and *mrpk*.
- **Distance proxy.** Distance is not perfect, but it is grounded in the literature on headquarters oversight and information frictions. It is also measured independently of the outcome variables.
- **Residualization strategy.** Firm-level moments are purged of industry, time, and size effects before being related to distance. This makes the model-data comparison cleaner.
- **Non-targeted validation.** The empirical variance of demeaned within-firm *mrpk* lies inside the model-implied range, which is reassuring because this moment was not directly targeted.

4.3 Main limitations

- **Distance is only a proxy.** Geographic spread may capture more than internal-capital-market frictions, including demand or input heterogeneity across locations.
- **Two-plant simplification.** Real firms often operate many plants. The mechanism should generalize, but the exact quantitative magnitudes may differ.
- **Strong assumption in cross-country exercise.** The India/China application assumes *mrpl* dispersion changes with firm structure in ways similar to *mrpk*, which may overstate or understate the true effect.
- **Interpretation of dispersion remains context dependent.** The paper does not claim that all within-firm dispersion is good; it claims that observed dispersion is the sum of good and bad components.

5 Tables and figures

5.1 Table 1: Cross-sectional moments of capital and investment

Read:

- The distribution of log *mrpk* is very dispersed: standard deviation 0.905 and interdecile range 2.120.
- The distribution is positively skewed, which is useful for thinking about why some plants sit far above the mean.
- Investment rates are extremely dispersed and leptokurtic, confirming that plant-level capital adjustment is lumpy rather than smooth.

- This table motivates fixed adjustment costs and the basic idea that large measured wedges in capital returns are common in the data.

TABLE 1
Cross-sectional moments of capital and investment

Variable	Mean	StD	IDR	Skewness	Kelley Skewn.	Excess kurtosis
$mrpk$		0.905 (0.013)	2.120 (0.032)	0.634 (0.028)	0.128 (0.010)	1.978 (0.085)
i/k	0.112 (0.015)	0.362 (0.093)	0.175 (0.008)	6.113 (0.099)	0.479 (0.008)	57.204 (2.378)

Notes: Data consist of our benchmark panel comprising annual plant-level data from the ASM 1972–2009. Moments are computed in a given year and 4-digit NAICS industry first before being aggregated by industry and then averaged across years. For details see Appendix A.4.

TABLE 2
Share of variance within firms in moments of capital and investment

Variable	A. Full MUF panel	B. Single-product firms
$mrpk$	0.601	0.559
i/k	0.679	0.689

Notes: The data underlying Panel A are our benchmark panel comprising annual plant-level data from the ASM 1972–2009. Moments in Panel B are based on a subsample of firms in which all plants produce the same and a single at the 7-digit SIC product level as defined by Foster *et al.* (2008). Moments are computed for each industry and years first before being aggregated by industry and then averaged across years. For details, see Appendix A.4.

5.2 Table 2: Share of variance within firms

Read:

- About 60.1% of $mrpk$ dispersion and 67.9% of investment-rate dispersion occur within firms.
- Restricting to single-product firms hardly changes the result: within-firm shares remain 55.9% and 68.9%.
- So the paper’s puzzle is not a compositional artifact from diversified conglomerates. The within-firm dimension is genuinely first order.

5.3 Figure 1: Frictions, output, and dispersion

Read:

- The figure presents the conceptual idea graphically: low frictions can coexist with high dispersion, while high frictions can coexist with low dispersion.
- The bottom panel emphasizes the non-monotonicity: output falls with frictions, but dispersion need not move one-for-one with efficiency.
- This is the figure that overturns the usual “more dispersion means more misallocation” intuition.

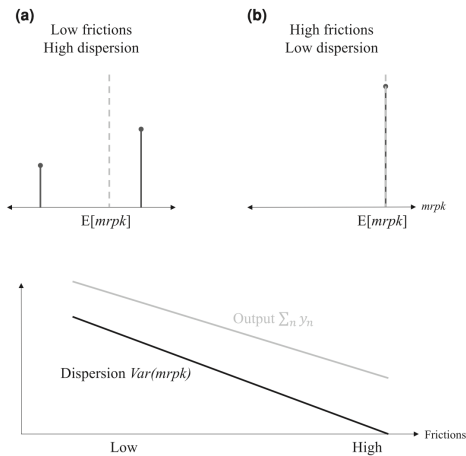


FIGURE 1

Frictions, output, and dispersion for low levels of frictions, depicted in Panel (a), and high levels of frictions, depicted in Panel (b).

TABLE 3 Model calibration			
Model param. and meaning	Value	Target/Source	
β	0.95	Long-run real interest rate	
a	0.627	ASM: estimated for multi-unit firms	
δ	0.067	ASM: long-run average i/k	
ρ^P	0.60	ASM: serial corr. of $\log(mrpk)_t$: 0.25	
ρ^f	0.85	ASM: serial corr. of $\log(mrpk)_t$: 0.31	
σ^P	0.25	ASM: volat. of $\log(mrpk)_t$: 0.33	
σ^f	0.24	ASM: volat. of $\log(mrpk)_t$: 0.26	
ψ_1	0.039	ASM: Cooper/Haltiwanger (2006)	
ψ_2	0.049	ASM: Cooper/Haltiwanger (2006)	
ζ_1	0.00154	ASM: Frequency of equity issuance of SUF	
ζ_2	0.11	ASM: Average size of ϕ in SUF	
γ	[0,1]	Vary between FullCM and NoICM	

5.4 Table 3: Calibration

Read:

- The plant production elasticity is 0.627 and depreciation is 0.067.
- The fixed and convex adjustment costs are quantitatively meaningful, as are the external-finance cost parameters.
- The internal-friction parameter γ is not fixed by calibration; it is the policy/friction object that the authors vary from 0 to 1 to compare FullICM with NoICM.

5.5 Figure 2: Aggregate effects of internal-capital-market frictions

Read:

- As γ rises, aggregate output and aggregate capital decline monotonically.
- But within-firm expected *mrpk* dispersion also declines. That is the central quantitative result of the paper.
- Moving from $\gamma = 0$ to $\gamma = 1$ lowers capital by about 12%, output by about 8%, and within-firm expected *mrpk* dispersion by more than 40%.
- Hence lower dispersion here is not a sign of better allocation; it comes from shutting down an internal margin of reallocation.

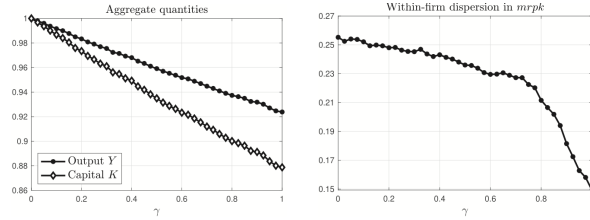


FIGURE 2
Aggregate quantitative effects of internal capital market frictions

Notes: This figure displays the quantitative effects of the internal capital market friction γ on dispersion and aggregates. Aggregate magnitudes are normalized to unity for the *FullICM* economy ($\gamma = 0$). Without ICM ($\gamma = 1$), aggregate capital and output respectively drop by 12% and 8% compared to the frictionless benchmark. At the same time, despite the additional friction, firms make investment choices that generate *lower* planned *mrpk* dispersion across their plants than in an economy populated by firms with fully functioning ICMs.

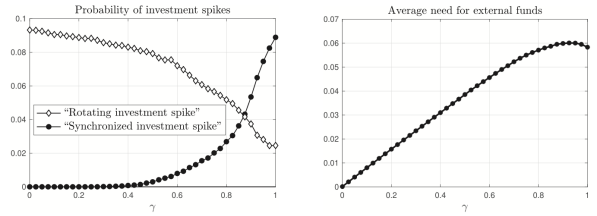


FIGURE 3
Quantitative effects of financial constraints on the multi-plant firm economy

Notes: This figure plots various moments of investment and the need for external funds and this associated costs as a function of the internal friction γ .

5.6 Figure 3: Quantitative mechanism inside the model

Read:

- With frictionless internal capital markets, rotating investment spikes are common; with no internal capital markets, synchronized spikes become common.
- The average need for external funds rises sharply with γ , reaching about 6% of capital in the NoICM economy.
- This figure makes the mechanism transparent: firms use internal capital markets to substitute away from costly external finance by staggering plant-level investment.

5.7 Table 4: Summary statistics of within-firm distance

Read:

- The average log-distance measure is 4.70 with substantial variation across firms.
- There is enough cross-sectional dispersion in distance to make it a meaningful empirical proxy for internal-capital-market frictions.
- This table is the bridge between the structural parameter γ in the model and observable firm heterogeneity in the data.

TABLE 4 Summary statistics of within-firm distance	
	Moment
Mean	4.70
St. Dev.	2.34
Skewness	-0.96

Source: Authors' calculations of the average distance across plants within firms in the ASM panel.

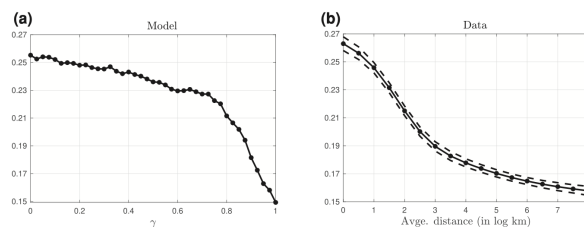


FIGURE 4
Dispersion of *mrpk* within firms

Notes: Panel (a) on the left: Dispersion of expected *mrpk* within firms in the quantitative model as in Figure 2. Panel (b) on the right plots its best empirical analogue, the variance of average revenue products of capital across plants within firms vis-a-vis geographic distance of the plants from the firm's centre point. Underlying sample comprises the set of 2-plant firms in the ASM.

5.8 Figure 4: Dispersion of *mrpk* within firms

Read:

- In the model, within-firm expected *mrpk* dispersion falls as γ rises.
- In the data, within-firm variance of revenue products is lower for geographically spread-out firms and higher for geographically concentrated firms.
- The sign and rough magnitude line up well with the model. Firms that likely have better internal capital markets show *more* dispersion, not less.

5.9 Figure 5: Firm-level *mrpk*

Read:

- In the model, average firm *mrpk* rises with γ because internal frictions make the firm more distorted overall.
- In the data, firms with more distant plants also exhibit higher firm-level *mrpk*.
- The combination of Figures 4 and 5 is crucial: good dispersion means higher variance in *mrpk* together with a *lower* level of *mrpk*.

5.10 Figure 6: External finance conditional on using external finance

Read:

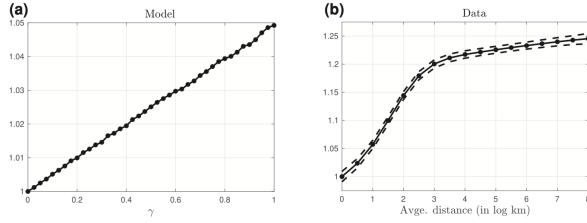


FIGURE 5
Firm-level $mrpk$

Notes: See notes to Figure 4.

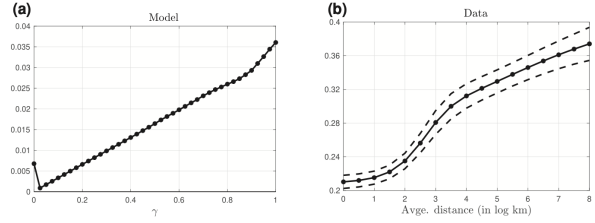


FIGURE 6
Volume of external financing conditional on external finance

Notes: See notes to Figure 4.

- In the model, the conditional size of external funding rises from near 0 to about 30% of capital as γ goes from 0 to 1.
- In the data, firms with more distant plants raise more external finance when they tap outside funds.
- This is direct evidence that poor internal capital markets push firms toward more expensive outside financing.

5.11 Figure 7: Probability of synchronized investment spikes

Read:

- The model predicts more synchronized plant-level spikes when internal capital markets are dysfunctional.
- The data show a broadly similar pattern: firms with plants farther apart are more likely to have synchronized spikes.
- This is exactly what one would expect if firms without good internal capital markets cannot rotate investment across plants efficiently.

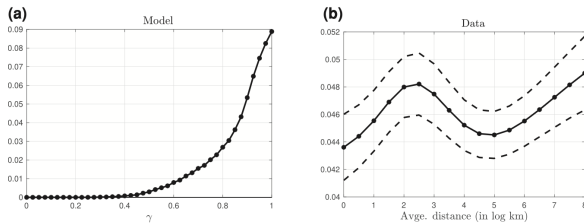


FIGURE 7
Probability of synchronized investment spikes

Notes: See notes to Figure 4.

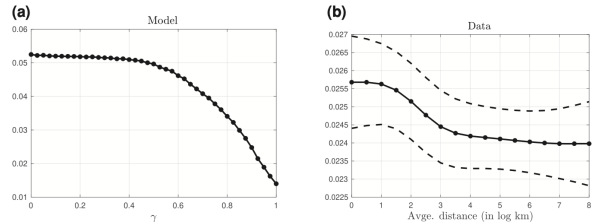


FIGURE 8
Within-firm co-movement of investment $Var(\Delta I/k)$

Notes: See notes to Figure 4.

5.12 Figure 8: Within-firm co-movement of investment changes

Read:

- In the model, $Var(\Delta I/k)$ is higher when γ is low because staggered investment creates plant-level asynchrony.
- In the data, geographically concentrated firms also show higher within-firm variance of investment-rate changes.

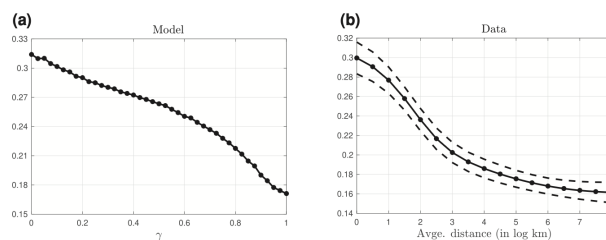


FIGURE 9
Within-firm co-movement of $mrpk$: $Var(\Delta mrpk)$

Notes: See notes to Figure 4.

- The economic reading is that better internal capital markets encourage reallocation over time rather than synchronization across plants.

5.13 Figure 9: Within-firm co-movement of revenue-product changes

Read:

- The model predicts higher $Var(\Delta mrpk)$ when internal capital markets are stronger.
- The data confirm the same pattern: concentrated firms show almost twice the variance of revenue-product changes as the most spread-out firms.
- This is a strong reduced-form confirmation of the staggered-investment mechanism because it links plant-level capital adjustment directly to plant-level return dynamics.

5.14 Section 5 exercise: revisiting Hsieh–Klenow

Read:

- The paper's cross-country exercise is not the core empirical contribution, but it is conceptually important.
- Once one recognizes that the U.S. contains good dispersion generated by multi-plant firms, poor countries should not be judged against raw U.S. dispersion levels.
- The paper's illustrative calculations imply larger gains from eliminating bad dispersion than previously thought: about 54.5% instead of 47% for India, and 57.4% instead of 39% for China.
- The exact numbers are preliminary, but the direction of the correction is the main point.

6 Short summary

- **Conclusion.** The paper shows that lower dispersion is not always better. Some within-firm dispersion is the by-product of efficient second-best reallocation through internal capital markets.
- **Mechanism.** Fixed investment costs make plant-level investment lumpy; costly external finance makes synchronized spikes expensive; internal capital markets let the firm stagger investment and save on outside finance.

- **Empirical lesson.** Within-firm geography helps reveal internal-capital-market frictions. Firms that are more geographically concentrated look more like the low- γ model economy.
- **Macro lesson.** Misallocation exercises that treat all observed dispersion as bad may understate the true gap between developed and emerging economies.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Dispersion is not monotone in frictions.** Relaxing a friction can increase output and increase dispersion at the same time.
- **Within-firm allocation is central.** Most measured dispersion in U.S. manufacturing occurs within firms, not across firms.
- **Internal capital markets matter quantitatively.** Eliminating frictions to internal reallocation raises capital and output materially.
- **Observed within-firm dispersion contains a good component.** A firm that reallocates funds well will often look *more* dispersed at the plant level because it deliberately staggers investment projects.
- **Cross-country implication.** Standard Hsieh–Klenow comparisons likely understate the potential gains from reallocation in economies with fewer multi-plant firms.

7.2 Economic intuition

- A firm faces a portfolio problem: it would like to invest where returns are high, but it also wants to avoid paying fixed adjustment costs at too many plants and external-finance costs at the firm level.
- A working internal capital market lets the firm concentrate funds on one plant today and another tomorrow. That raises cross-plant heterogeneity in capital returns in the short run.
- The resulting dispersion is not evidence that the firm failed to equalize returns. It is evidence that the firm is implementing the best feasible allocation under non-convex investment and financing frictions.

7.3 Takeaway

This paper is important because it changes how one should read productivity dispersion in plant data. In standard misallocation work, dispersion is usually a reduced-form symptom of wedges. Kehrig and Vincent show that once firms are multi-plant and financially constrained, the same dispersion can arise from an efficient use of internal capital markets. The practical lesson is simple but powerful: before treating dispersion as a sufficient statistic for misallocation, one must understand the organization of the firm and the margins through which it reallocates resources internally.